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Investigating the influences of Fe, Mn and Mo additions on the evolution of microstructure and mechanical performances of Al–Si–Mg cast alloys

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ABSTRACT

Understanding the evolution of Fe-containing intermetallic compounds (Fe–IMCs) in Al–Si–Mg alloys is essential for improving the mechanical performances. This paper aims to systemically evaluate the influences of both the single addition of Mn or Mo, and the combined additions of Mn and Mo on the evolution of Fe–IMCs and mechanical performances of Al–Si–Mg cast alloys. Results reveal that the as-cast microstructures of these alloys are primarily composed of Al₃M (M means Ti, Zr, or TiZr), α -Al, eutectic Si, Mg₂Si, Q–Al₅Cu₂Mg₈Si₆, and some irregular Fe–IMCs. It is clear that the morphology and size of Fe–IMCs are significantly affected by the single or combined addition of Mn and Mo. Moderate Mn content facilitates the formation of Chinese script-like α -Al(FeMn)Si phases that are favorable to the mechanical properties. The combined addition of Mn and Mo refines the α -Al(FeMn)Si phases and promotes the formation of Chinese script-like α -Al(FeMnMo)Si phases. Furthermore, as the Mo level increases, β -AlFeSi, π -Al₈FeMg₃Si₆, and α -Al(FeMn)Si disappear successively, which also helps improve the mechanical performances. In this article, maximum hardness, ultimate tensile strength, yield strength and elongation of 87.04 HV, 275.34 MPa, 139.70 MPa, 7.88% are obtained when Fe, Mn and Mo are added at 0.20wt.%, 0.02wt.%, and 0.27wt.%, respectively. The purpose of this work is to provide an efficient method to minimize or even eliminate the adverse effects of Fe–IMCs and thus improve the mechanical performances of Al–Si–Mg cast alloys.

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1. Introduction

With the ever-increasing requirements of automobile lightweight, there is an urgent need to develop a high-performance material to replace traditional steel constructions in vehicle designs [1]. Nowadays, one of the most popular materials for the lightweight design of automobiles, Al–Si–Mg cast alloys have been broadly applied in the field of automobile structural parts such as car body structures, wheels, gearboxes, and battery housings, due to their distinct characteristics such as low density, easy castability, excellent weldability, good corrosion resistance and suitable die-casting design [2–8]. Unfortunately, Al–Si–Mg cast alloys are also highly susceptible to some brittle Fe-containing intermetallic compounds (Fe–IMCs) after the use of Si modification and α -Al grain refinement [4,9–11]. These unfavorable Fe–IMCs will lead to inferior mechanical properties, especially poor tensile strength and ductility, which largely limits their widespread applications in the automotive sector.

As far as Al–Si–Mg cast alloys are concerned, the most common intermetallic compounds (IMCs) identified during solidification are generally β -Al₅FeSi (β -Fe) phases, α -Al₁₅(-FeMn)₃Si₂ (α -Fe) phases, and π -Al₈FeMg₃Si₆ (π -Fe) phases, Q-Al₅Cu₂Mg₈Si₆ phases, and so on [12–16]. Among them, the α -Fe phases and β -Fe phases have been documented to largely influence the castability and mechanical properties of the alloy. The amount, size and morphology of these Fe–IMCs are largely depend on the casting process, solidification rates and alloy compositions, especially the Fe contents [13]. It is now well accepted that the coarse needle/plate-like β -Fe phases are the most harmful one for tensile properties, especially ductility, as they are potential sites for crack initiation [17,18]. The coarse needle/plate-like β -Fe phases may also block the interdendritic melt flow, resulting in shrinkage cavities in the final components [19]. The large-sized hard and brittle α -Fe phases also adversely affect the mechanical performances, while the Chinese script-like ones are less harmful [12]. Moreover, the Chinese script-like π -Fe phases in the Al–Si–Mg cast alloys are easily decomposed into thin needle/plate-like β -Fe phases after solution treatment, which no further improves the mechanical properties of the alloy [14]. Therefore, it is really necessary to lower the hazard of these undesirable Fe–IMCs in Al–Si–Mg cast alloys.

Under this circumstance, reasonably controlling the adverse effects of these harmful Fe–IMCs, especially the needle/plate-like β -Fe phases, is of great significance to enhance the mechanical performance of this alloy [20–22]. Many scientists have devoted a lot of energy to this. Among them, two main ideas prevail. One is to physically remove or separate the Fe–IMCs via gravity sedimentation, centrifugation or magnetic separation, and the other is to chemically modify the morphology of Fe–IMCs from needle/plate-like into a more compact Chinese script-like [12,23,24]. Currently, the removal or separation efficiency of β -Fe phases is insufficient, while its morphology change is easier to achieve. It has been demonstrated that, both imposing rapid cooling rate and adding transition metals (TM) such as Mn, Cr, and Co in the melt are effective methods to modify the undesired brittle Fe–IMCs [13,23–28]. Up to now, chemical modification

methods of brittle Fe–IMCs are still very popular. For example, Mn can effectively neutralize the adverse effects of needle/plate-like β -Fe phases by replacing Fe to generate less harmful Chinese script-like α -Fe phases, thereby significantly enhancing the tensile strength and ductility [12,23]. In addition, Mn can also reduce the embrittlement action by forming Chinese script-like phases [13,29–31]. Most of the previous reports on Mn-modified brittle Fe–IMCs are mainly devoted to determining the optimal ratio of Mn/Fe [32]. However, the influence of Mn/Fe ratios on the brittle Fe–IMCs morphology remains controversial. Whether the Mn/Fe ratio is too high or too low, the satisfactory Fe–IMCs cannot be achieved in the microstructure since their morphology, size, and brittleness are adverse. Moreover, Mo has been identified as an excellent alternative to Mn, and has received much attention [33,34]. Regrettably, there are few systematical studies about the influences of both the single addition of Mn or Mo, and the combined additions of Mn and Mo on the precipitation behavior and the evolution of Fe–IMCs in Al–Si–Mg cast alloys.

In this work, the evolution of Fe–IMCs during the solidification of Al–Si–Mg cast alloys with different Fe, Mn, and Mo levels are systemically investigated. Through the statistical analysis, the critical effects of Fe, Mn, and Mo on the Fe–IMCs are identified and their impacts on mechanical performances of Al–Si–Mg cast alloys are further reported. Moreover, the solidification sequence, precipitation temperature, chemical composition and mass fraction of Fe–IMCs are explored by thermodynamic calculation, so that their amount, size and morphology could be effectively controlled. This provides a highly effective method to minimize or even eliminate the adverse effect of brittle Fe–IMCs, and we believe that our research will stimulate new thinking.

2. Experimental

2.1. Alloy preparation procedures

In our work, Al–Si–Mg cast alloys were used as the raw material. This alloy was prepared from pure Al (99.99 wt.%), Mg (99.9 wt.%), Cu (99.9 wt.%), Zn (99.9 wt.%), and Al–20Si, Al–20Fe, Al–10Mn, Al–5Mo, Al–10Ti, Al–10Zr and Al–10Sr master alloys (all in wt.% unless otherwise stated), respectively. The alloy preparation procedures were shown in Fig. 1. The fusion of all pure metals and master alloys (2 Kg in total) was first performed in graphite crucibles at the specified melting temperature (720 °C) using an intermediate frequency furnace. The melt was then modified by the addition of Sr (150 ppm) just 3 min after degassing with hexachloroethane and held at 720 °C for 10min. The melt was finally transferred from the furnace to a water-cooled copper mold (cooling rate: about 500 K/s) at 720 °C. The actual chemical compositions of the alloy used in the experiment were determined by direct-reading spectrometer (SPECTRO LAB), and the results were listed in Table 1. To reduce the negative effect of more Fe–IMCs on mechanical performances of Al–Si–Mg cast alloys, the sum of Fe, Mn and Mo concentrations (0.5 wt.%) designed should be kept as consistent as possible.

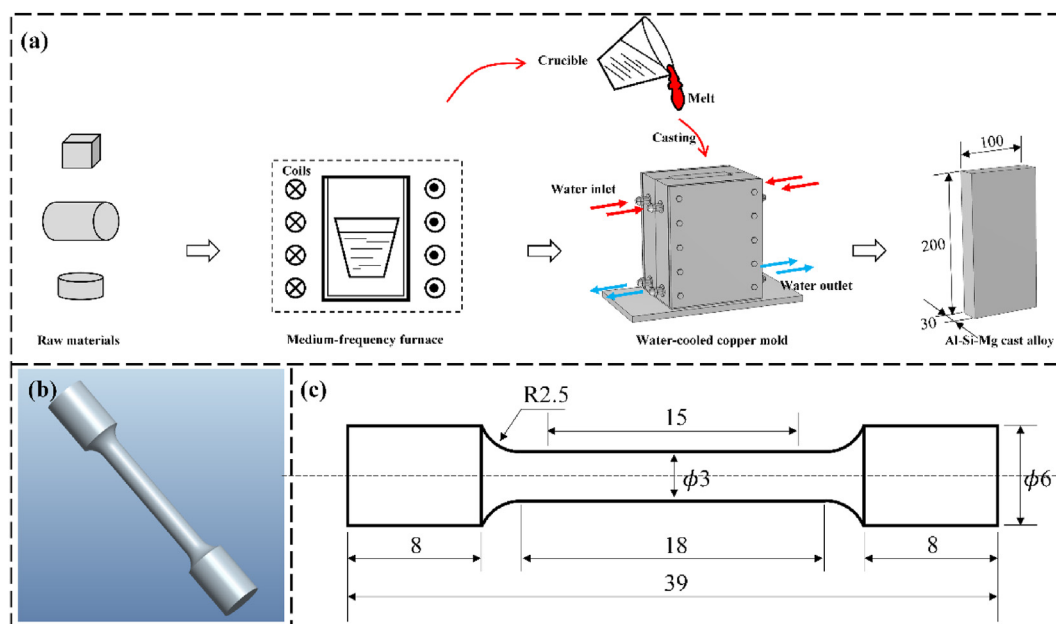


Fig. 1 – (a) Schematic of the alloy casting process and (b) tensile bars; (c) the sampling size of the tensile bars (unit: mm).

2.2. Thermodynamic calculation

To investigate the influences of both the single addition of Mn or Mo, and the combined additions of Mn and Mo on the precipitation behavior of Al–Si–Mg cast alloys with various Fe contents during solidification, the thermodynamic calculations were performed using JMatPro software. The non-equilibrium solidification paths of Al–Si–Mg cast alloys were determined by using the Scheil model. The influences of Fe, Mn and Mo levels on the solidification sequence, temperature, chemical composition and mass fraction of Fe–IMCs were mainly examined.

2.3. Microstructural characterization

Using standard metallographic methods, all samples were first ground using SiC papers, and next polished with (1.5 μm) diamond abrasives. The microstructures were captured by using an OLYMPUS BX53 optical metallographic microscope (OM) and a TESCAN MIRA scanning electron microscope (SEM) equipped with energy dispersive spectrometer (EDS), with an accelerating voltage of 20 kV. To examine and quantify the

evolution of Fe–IMCs in the experimental alloys, the average length and width (or diameter), as well as area fraction of various Fe–IMCs were quantitatively measured by Image-Pro Plus 6.0 image analysis software. Among them, the formula involved is as follows,

$$d = 2 \times \sqrt{A/\pi} \quad (1)$$

where d and A represent the average grain size (μm) and the area (μm^2) of a single Fe–IMCs, respectively. To ensure the reliability of the data, at least 30 independent fields at 500 times magnification (approximately $225,185.5 \mu\text{m}^2$) were randomly selected on each sample surface.

2.4. Mechanical properties

Tensile tests (at room temperature) were performed using a general electronic tensile testing apparatus (CSS–44100 type). At least three standard tensile bars (were machined parallel to the casting direction of cast ingots according to GB/T 16,865–2013) were tested at a strain rate of 1 mm/min, and their dimensions are exhibited in Fig. 1. The tensile fracture morphology was investigated using a TESCAN MIRA3 type

Table 1 – The actual chemical composition of Al–Si–Mg cast alloys used in the experiment (wt.%).

Alloy	Actual chemical compositions (wt.%)										
	Si	Fe	Mn	Mo	Cu	Mg	Ti	Zn	Zr	Sr	Al
A1 (base alloy)	9.05	0.11	0.41	0	0.30	0.40	0.085	0.19	0.15	0.012	Bal.
A2	8.89	0.19	0.29	0	0.31	0.38	0.076	0.18	0.14	0.014	Bal.
A3	8.85	0.33	0.21	0	0.32	0.39	0.083	0.17	0.14	0.013	Bal.
A4	8.68	0.21	0.19	0.09	0.29	0.41	0.088	0.18	0.13	0.012	Bal.
A5	8.79	0.22	0.11	0.19	0.34	0.37	0.079	0.17	0.15	0.012	Bal.
A6	8.99	0.20	0.02	0.27	0.32	0.38	0.082	0.16	0.17	0.011	Bal.

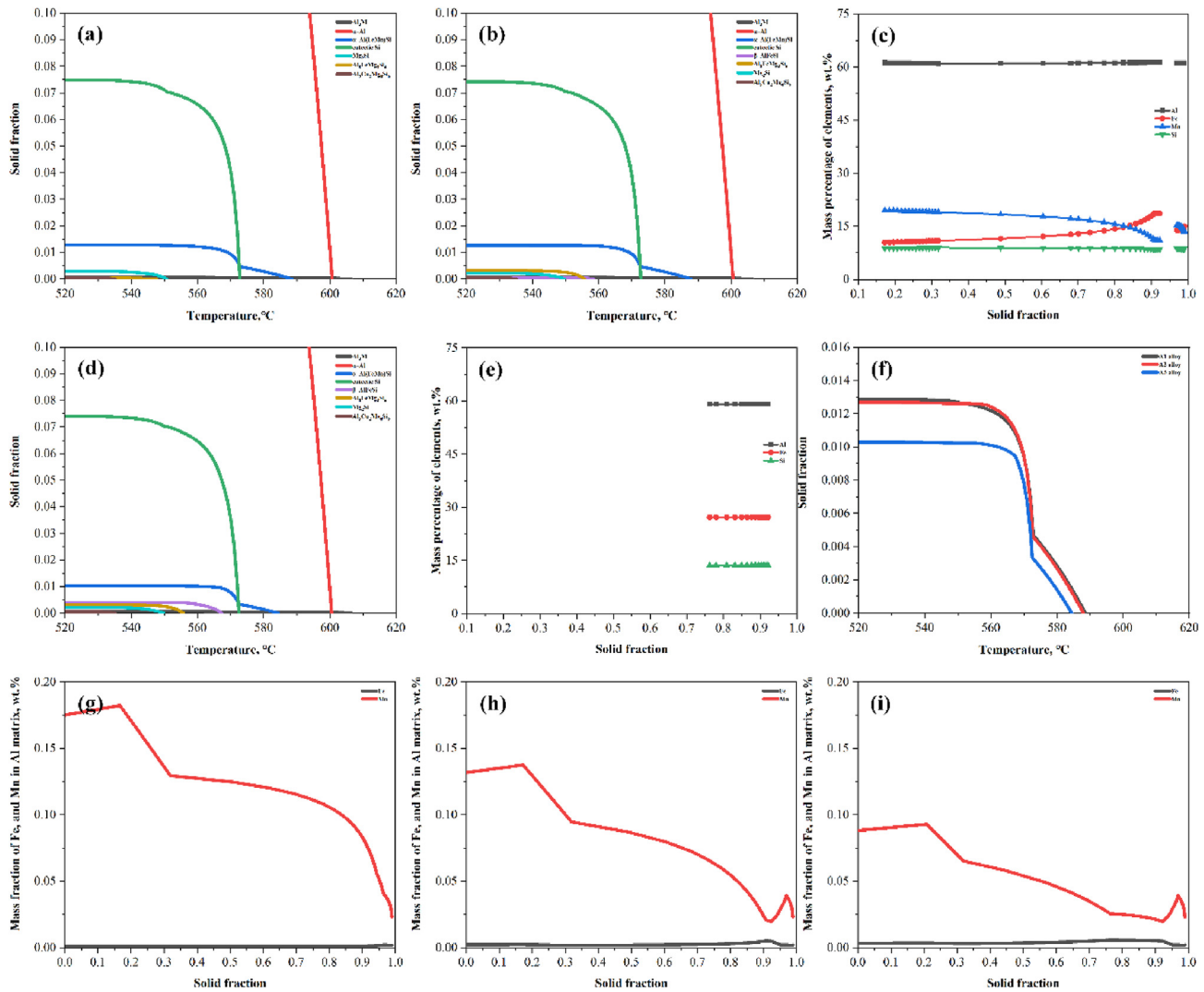


Fig. 2 – (a) Calculated solidification sequence of A1 alloy; (b) calculated solidification sequence of A2 alloy; (c) composition of α -Al(FeMn)Si in A2 alloy; (d) calculated solidification sequence of A3 alloy; (e) composition of β -AlFeSi in A3 alloy; (f) precipitation temperatures of α -Al(FeMn)Si with different Mn/Fe ratios; mass fraction curves of Fe and Mn in the Al matrix of (g) A1, (h) A2, (i) A3 alloys.

SEM. Moreover, hardness measurements were carried out for 10s at a load of 25gf using a micro Vickers hardness tester (Wolpert Wilson). At least 9 indentations were made for each sample.

3. Results and discussion

3.1. Thermodynamic calculations

Figs. 2 and 3 depict the influences of both the single addition of Mn or Mo, and the combined additions of Mn and Mo on the solidification sequence, precipitation temperature, chemical composition and mass fraction of Fe-IMCs in Al-Si-Mg cast alloy with various Fe levels. Fig. 2(a–b, d) and Table 2 display the calculated solidification sequence of A1, A2 and A3 alloys, respectively. It can be seen that, the precipitates are mainly

composed of Al_3M (M means Ti, Zr, or TiZr), α -Al, α -Al(FeMn)Si, eutectic Si, β -AlFeSi, Mg_2Si , $\text{Al}_8\text{FeMg}_3\text{Si}_6$, and $\text{Al}_5\text{Cu}_2\text{Mg}_8\text{Si}_6$. Fig. 2(a) shows that, when the temperature is in the range of 520–620 °C, Al_3M (606 °C) and α -Al (600 °C) phases crystallize first, and then α -Al(FeMn)Si (589 °C) and eutectic Si (573 °C) phases separate out in A1 alloy. With the decrease of temperature, Mg_2Si (550 °C) and $\text{Al}_8\text{FeMg}_3\text{Si}_6$ (547 °C) phases are also precipitated in turn, followed by a small amount of $\text{Al}_5\text{Cu}_2\text{Mg}_8\text{Si}_6$ (535 °C) phase. Fig. 2(b) presents that, with the decrease of temperature in the range of 520–620 °C, Al_3M , α -Al, α -Al(FeMn)Si, eutectic Si, β -AlFeSi, $\text{Al}_8\text{FeMg}_3\text{Si}_6$, Mg_2Si and $\text{Al}_5\text{Cu}_2\text{Mg}_8\text{Si}_6$ phases are successively precipitated in A2 alloy. Among them, the α -Al(FeMn)Si phase precipitates at 588 °C, and its composition (62% Al, 15% Fe, 14%Mn and 9% Si) is shown in Fig. 2(c). Unlike A1 alloy, a small amount of β -AlFeSi phase appears for the first time at 559 °C in A2 alloy. Moreover, the solidification sequence of A3 alloy (Fig. 2(d)) is

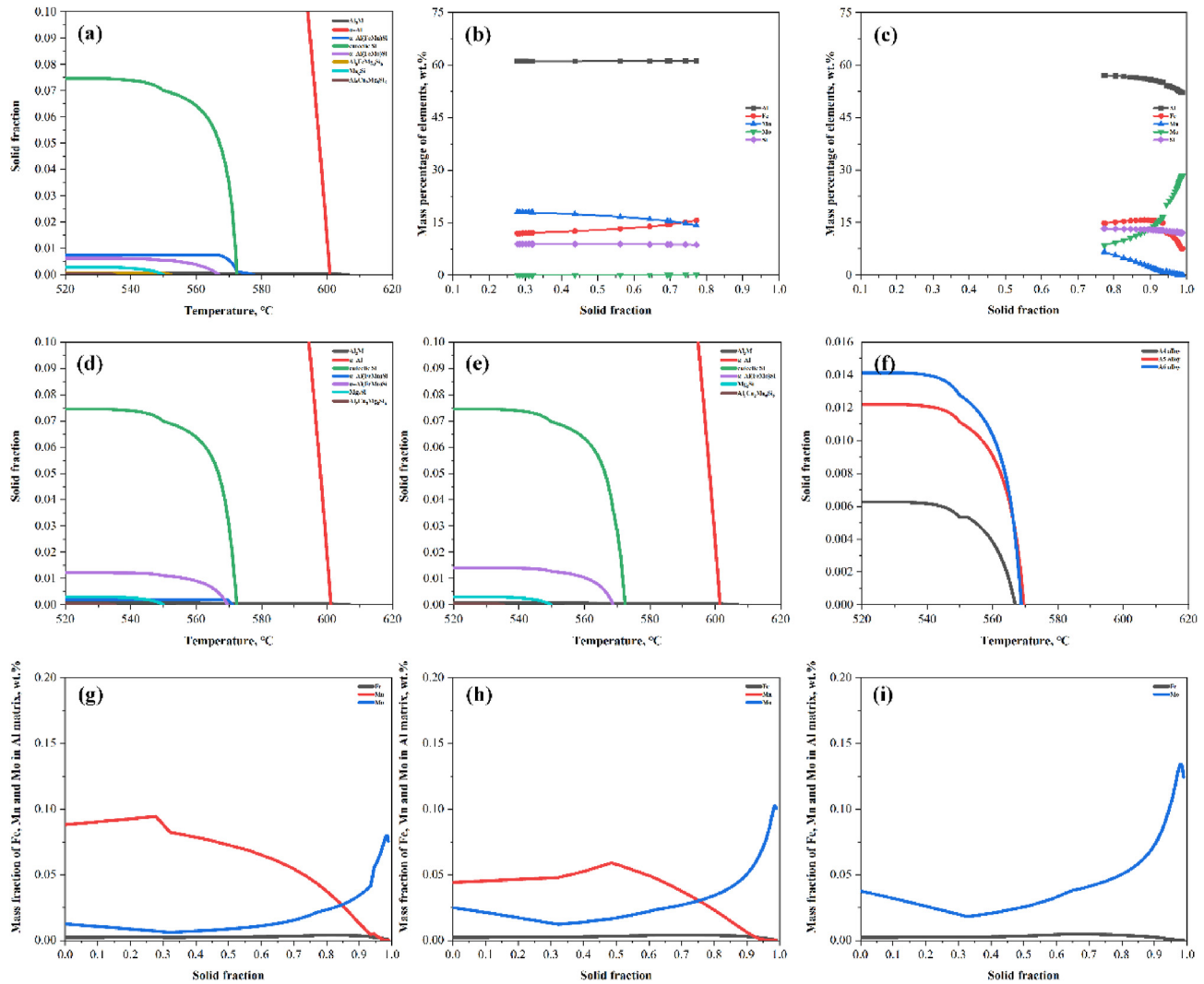


Fig. 3 – (a) Calculated solidification sequence of A4 alloy; composition of (b) α -Al(FeMn)Si and (c) α -Al(FeMo)Si in A4 alloy; (d) calculated solidification sequence of A5 alloy; (e) calculated solidification sequence of A6 alloy; (f) precipitation temperatures of α -Al(FeMo)Si with different Mo additions; mass fraction curves of Fe, Mn and Mo in the Al matrix of (g) A4, (h) A5, (i) A6 alloys.

consistent with that of A2 alloy. In particular, the amount of β -AlFeSi phase increases, its precipitation temperature is 568 °C, and the composition (59% Al, 27% Fe and 14% Si) is given in Fig. 2(e). Fig. 2(f) indicates the precipitation temperature of the α -Al(FeMn)Si phase with different Mn/Fe ratios. Clearly, the Mn/Fe ratio leads to a significant change in the precipitation temperature of α -Al(FeMn)Si phases. With the decrease of the Mn/Fe ratio, the precipitation temperature of the α -Al(FeMn)Si phase gradually reduces, although the sum of Fe and Mn contents is constant. Moreover, the precipitation temperature of α -Al(FeMn)Si phases may also be related to its total amount. This is confirmed by the comparison of the precipitation temperatures of A2 and A3 alloys. Fig. 2(g–i) describes the Fe and Mn mass fraction curves in the Al matrix, respectively. It can be seen that, in these three alloys (A1, A2, A3), on the one hand, as the solid fraction increases, the Mn mass fraction in the Al matrix both first increases slowly and

then decreases sharply, while the Fe mass fraction in the Al matrix does not change much and remains at a very low level. On the other hand, almost all Mn participate in the formation of α -Al(FeMn)Si phases, but the neutralization ability of Mn to Fe is weakened due to the decreased Mn level, resulting in insufficient modification of Fe. The residual Fe level forms the β -AlFeSi and/or $\text{Al}_8\text{FeMg}_3\text{Si}_6$ phases. In addition, the total amount of Fe–IMCs is nearly the same because the mass fraction of (Fe + Mn) in the Al matrix is constant.

Fig. 3(a, d–e) and Table 2 exhibit the calculated solidification sequence of A4, A5 and A6 alloys, respectively. It can be found that, the precipitates are primarily made up of Al_3M (M means Ti, Zr, or TiZr), α -Al, α -Al(FeMn)Si, eutectic Si, α -Al(FeMo)Si, $\text{Al}_8\text{FeMg}_3\text{Si}_6$, Mg_2Si , $\text{Al}_5\text{Cu}_2\text{Mg}_8\text{Si}_6$. Compared with A2 alloy, the precipitates that appear in A4 alloy during the solidification process do not actually change much, except that the β -AlFeSi phase disappeared and the new α -Al(FeMo)

Table 2 – Predicted solidification sequence (in the range of 520–620 °C) of six Al–Si–Mg cast alloys.

Alloy	Temperature, °C	Phase	Solidification sequence
A1	606	Al ₃ M	1: L → Al ₃ M
	600	α-Al	2: L → Al ₃ M + α-Al
	589	α-Al(FeMn)Si	3: L → Al ₃ M + α-Al + α-Al(FeMn)Si
	573	eutectic Si	4: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si
	550	Mg ₂ Si	5: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + Mg ₂ Si
	547	π-Al ₈ FeMg ₃ Si ₆	6: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + Mg ₂ Si + π
	535	Q-Al ₅ Cu ₂ Mg ₈ Si ₆	7: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + Mg ₂ Si + π + Q
A2	606	Al ₃ M	1: L → Al ₃ M
	600	α-Al	2: L → Al ₃ M + α-Al
	588	α-Al(FeMn)Si	3: L → Al ₃ M + α-Al + α-Al(FeMn)Si
	573	eutectic Si	4: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si
	559	β-AlFeSi	5: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi
	556	π-Al ₈ FeMg ₃ Si ₆	6: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi + π
	550	Mg ₂ Si	7: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi + π + Mg ₂ Si
	535	Q-Al ₅ Cu ₂ Mg ₈ Si ₆	8: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi + π + Mg ₂ Si + Q
A3	606	Al ₃ M	1: L → Al ₃ M
	600	α-Al	2: L → Al ₃ M + α-Al
	584	α-Al(FeMn)Si	3: L → Al ₃ M + α-Al + α-Al(FeMn)Si
	573	eutectic Si	4: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si
	568	β-AlFeSi	5: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi
	556	π-Al ₈ FeMg ₃ Si ₆	6: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi + π
	550	Mg ₂ Si	7: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi + π + Mg ₂ Si
	535	Q-Al ₅ Cu ₂ Mg ₈ Si ₆	8: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + β-AlFeSi + π + Mg ₂ Si + Q
A4	606	Al ₃ M	1: L → Al ₃ M
	601	α-Al	2: L → Al ₃ M + α-Al
	578	α-Al(FeMn)Si	3: L → Al ₃ M + α-Al + α-Al(FeMn)Si
	573	eutectic Si	4: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si
	567	α-Al(FeMo)Si	5: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + α-Al(FeMo)Si
	552	π-Al ₈ FeMg ₃ Si ₆	6: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + α-Al(FeMo)Si + π
	550	Mg ₂ Si	7: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + α-Al(FeMo)Si + π + Mg ₂ Si
	535	Q-Al ₅ Cu ₂ Mg ₈ Si ₆	8: L → Al ₃ M + α-Al + α-Al(FeMn)Si + Si + α-Al(FeMo)Si + π + Mg ₂ Si + Q
A5	606	Al ₃ M	1: L → Al ₃ M
	601	α-Al	2: L → Al ₃ M + α-Al
	573	eutectic Si	3: L → Al ₃ M + α-Al + Si
	571	α-Al(FeMn)Si	4: L → Al ₃ M + α-Al + Si + α-Al(FeMn)Si
	570	α-Al(FeMo)Si	5: L → Al ₃ M + α-Al + Si + α-Al(FeMn)Si + α-Al(FeMo)Si
	550	Mg ₂ Si	6: L → Al ₃ M + α-Al + Si + α-Al(FeMn)Si + α-Al(FeMo)Si + Mg ₂ Si
	535	Q-Al ₅ Cu ₂ Mg ₈ Si ₆	7: L → Al ₃ M + α-Al + Si + α-Al(FeMn)Si + α-Al(FeMo)Si + Mg ₂ Si + Q
A6	607	Al ₃ M	1: L → Al ₃ M
	601	α-Al	2: L → Al ₃ M + α-Al
	573	eutectic Si	3: L → Al ₃ M + α-Al + Si
	569	α-Al(FeMo)Si	4: L → Al ₃ M + α-Al + Si + α-Al(FeMo)Si
	550	Mg ₂ Si	5: L → Al ₃ M + α-Al + Si + α-Al(FeMo)Si + Mg ₂ Si
	535	Q-Al ₅ Cu ₂ Mg ₈ Si ₆	6: L → Al ₃ M + α-Al + Si + α-Al(FeMo)Si + Mg ₂ Si + Q

Si phase appeared (Fig. 3(a)). Among them, the compositions of α-Al(FeMn)Si and α-Al(FeMo)Si phases are displayed in Fig. 3(b and c). Fig. 3(b) displays that, the α-Al(FeMn)Si phase is mainly composed of 61.2% Al, 15.6% Fe, 14.2% Mn and 8.7% Si, accompanied by a small amount of Mo (<1%). Meanwhile, Fig. 3(c) presents that, the α-Al(FeMo)Si phase chiefly consists of 52.1% Al, 7.5% Fe, 28.2% Mo and 12.1% Si, and also includes a small amount of Mn (<1%). Obviously, the results indicate that both Mn and Mo can replace the Fe without changing the crystal structure, which is in agree with the previous literatures [34]. Compared with A4 alloy, the precipitates that occurred in A5 and A6 alloys have significantly changed during the solidification process (Fig. 3(d and e)). With the decrease of Mn addition and the increase of Mo addition, the amount of the α-Al(FeMn)Si phase gradually decreases, while the amount of the α-Al(FeMo)Si phase gradually increases.

Especially in the absence of Mn, only the α-Al(FeMo)Si phase is formed in A6 alloy. It can also be discovered that, the Al₈FeMg₃Si₆ phase disappears when the Mo addition is greater than 0.2 wt.%. This indicates that Mo neutralizes Fe better than Mn [33]. Furthermore, the Mo addition leads to a significant change in the precipitation temperature of the α-Al(FeMo)Si phase. It can be found from Fig. 3(f) that, as the Mo addition increases, the precipitation temperatures of the α-Al(FeMo)Si phase increase first and then decrease slightly. Fig. 3(g–i) describes the mass fraction curves of Fe, Mn and Mo in the Al matrix, respectively. It can be seen that, in these three alloys (A4, A5, A6), the Fe and Mn mass fractions in the Al matrix have a similar changing trend to that of A2 alloy, while the Mo mass fraction first decreases slowly and then increases significantly. The results indicate that, with the increase of Mo addition and the decrease of Mn addition, the

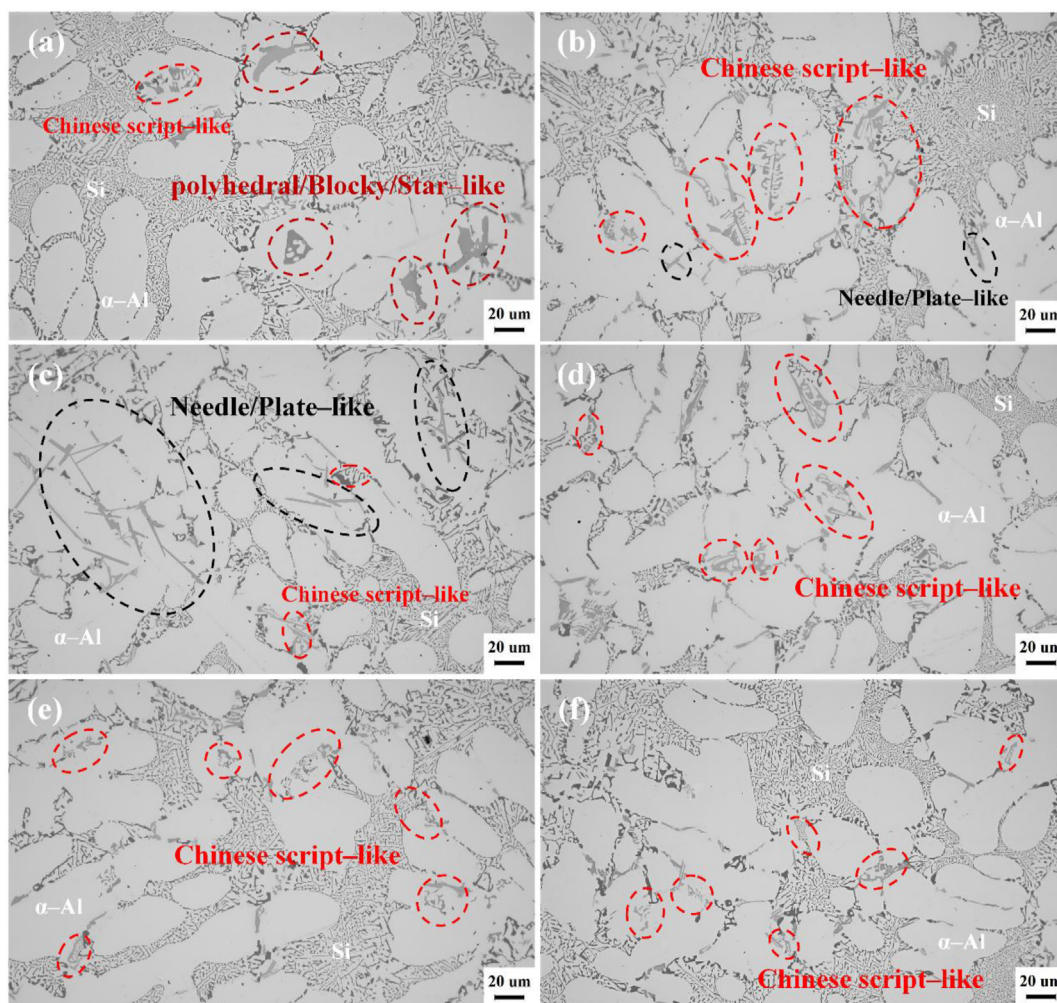


Fig. 4 – OM images of as-cast microstructures of Sr-modified Al–Si–Mg cast alloys with different Fe, Mn, and Mo additions. (a) A1 alloy (base alloy); (b) A2 alloy; (c) A3 alloy; (d) A4 alloy; (e) A5 alloy; (f) A6 alloy.

α -Al(FeMo)Si phase formed during solidification increases while the α -Al(FeMn)Si phase decreases. Furthermore, the sum of mass fractions of α -Al(FeMn)Si and α -Al(FeMo)Si phases slightly increases, although the (Fe + Mn + Mo) mass fractions in the Al matrix increases. In short, thermodynamic calculations are important for predicting the precipitation and preferential selection of Fe-IMCs in Al–Si–Mg cast alloys.

3.2. Microstructural characterization

Fig. 4 reflects the microstructures of Sr-modified Al–Si–Mg cast alloys with various Fe, Mn, and Mo additions. Fig. 4(a) presents that the typical as-cast microstructure of A1 alloy (base alloy) is primarily composed of coarse α -Al dendrites, fine eutectic Si particles, and some dark grey IMCs. These irregular IMCs are mainly polyhedral/blocky/star-like, and a few are Chinese script-like, which are mostly embedded in the Al matrix, or in contact with eutectic Si particles. With the decrease of the Mn/Fe ratio, the α -Al dendrites and eutectic Si particles in A2 and A3 alloy do not change obviously, while the dark grey phases change significantly, especially their shape,

size and distribution. Fig. 4(b) reveals that the more polyhedral/blocky/star-like dark grey phases in A2 alloy are replaced by a large number of coarse and compact Chinese script-like ones, which are mainly distributed at the modified eutectic grain boundaries. Moreover, a small amount of needle/plate-like dark grey phases begin to appear. Fig. 4(c) displays that a large number of needle/plate-like dark grey phases randomly appear in A3 alloy. The predominant needle/plate-like dark grey phases in the Al matrix coexist with some Chinese script-like ones. This clearly shows that the Mn/Fe ratio is of great significance for the phase transformation of the dominant dark grey phase, and with the increasing Mn/Fe ratio, the dominant dark grey phase changes from needle/plate-like to polyhedral/blocky/star-like. Furthermore, Fig. 4(d–f) shows the morphology of the dominant dark grey phases of A4 to A6 alloys with trace Mo additions is similar to that of the A2 alloy without Mo addition (Fig. 4(b)), mainly in Chinese script-like rather than needle/plate-like or polyhedral/blocky/star-like ones. And with the increase of Mo addition, the size and area fraction of Chinese script-like phases formed during solidification gradually decrease. This

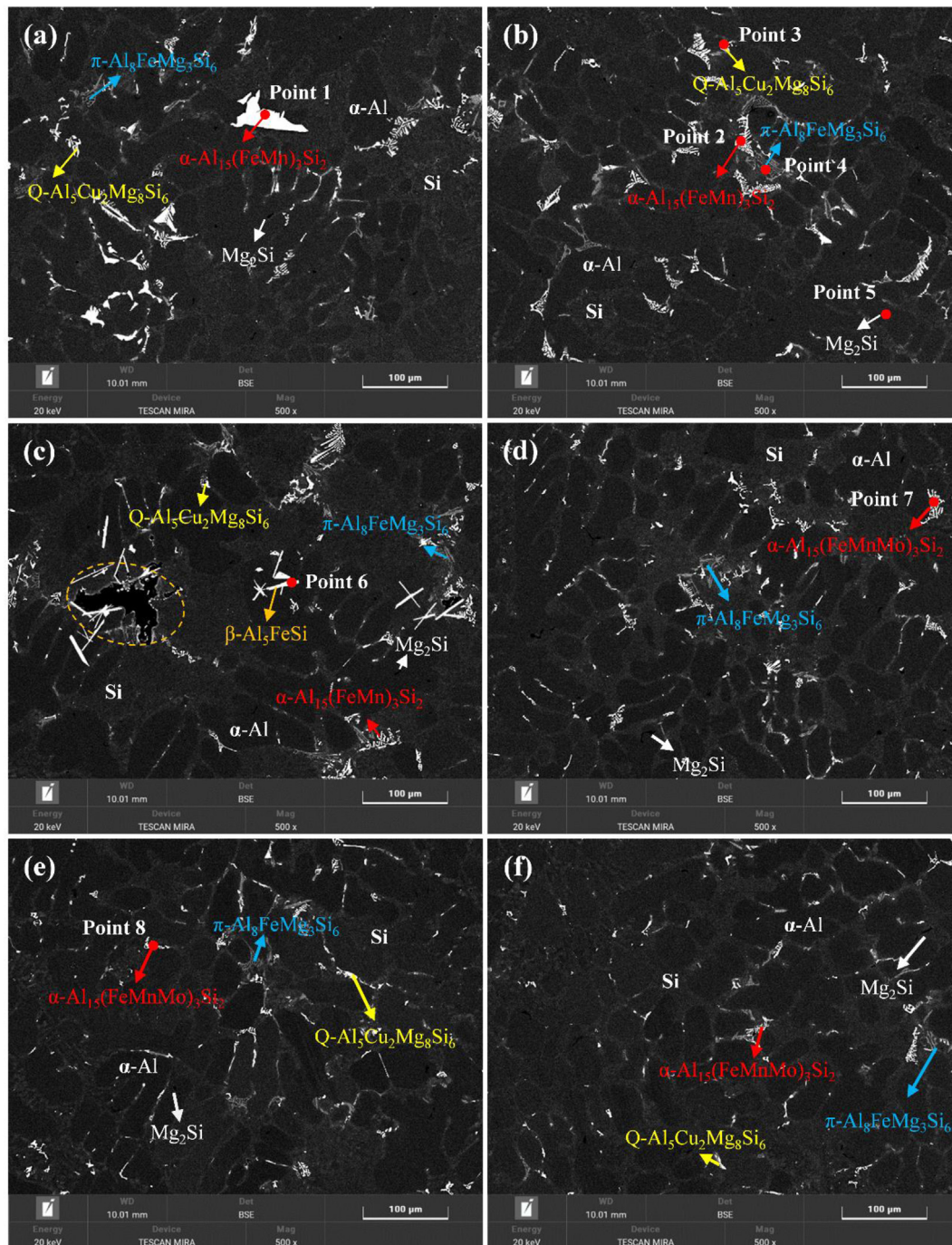


Fig. 5 – SEM images of the morphology evolution of IMCs in Al–Si–Mg cast alloys depending on the addition of Fe, Mn, and Mo. (a) A1 alloy (base alloy); (b) A2 alloy; (c) A3 alloy; (d) A4 alloy; (e) A5 alloy; (f) A6 alloy.

indicates that the addition of Mo has an important influence on the formation and growth of Chinese script-like phases in Al–Si–Mg cast alloys.

According to previous works, some meaningful IMCs with different morphological features may also be included in the microstructure of Al–Si–Mg cast alloys. For example, the β -Fe phase exhibits the needle/plate-like morphology, whereas the α -Fe phase, π phase, Q phase, and Mg_2Si phase are usually

polyhedral/blocky/star-like, Chinese script-like or skeleton-like morphologies [10,35,36]. However, they are indistinguishable in the OM microstructures. Among them, the most important and common IMCs are the α -Fe phase and the β -Fe phase, respectively [12]. The β -Fe phase may transform into the α -Fe phase after both the single addition of Mn or Mo, and the combined additions of Mn and Mo, since Fe is easily replaced by Mn and Mo during solidification [29,37]. Besides,

Table 3 – EDS point scanning element content (at.%) in Fig. 5.

Point	Elements (at.%)							Suggested formula
	Al	Si	Mg	Cu	Fe	Mn	Mo	
1	68.96	11.89	0.1	0	14.81	4.24	—	α -Al ₁₅ (FeMn) ₃ Si ₂
2	72.35	12.85	0.14	0.02	10.66	3.98	—	α -Al ₁₅ (FeMn) ₃ Si ₂
3	64.23	18.23	13.65	3.55	0.29	0	—	Q-Al ₅ Cu ₂ Mg ₈ Si ₆
4	55.36	26.4	14.23	0	4.01	0	—	π -Al ₈ FeMg ₃ Si ₆
5	59.36	15.33	25.1	0	0.11	0.1	—	Mg ₂ Si
6	70.41	16.31	0.17	0	13.1	0	—	β -Al ₅ FeSi
7	72.91	9.88	0.64	0	8.87	4.3	3.4	α -Al ₁₅ (FeMnMo) ₃ Si ₂
8	79.33	10.05	0.18	0.74	7.66	0	2.04	α -Al ₁₅ (FeMo) ₃ Si ₂

the β -Fe phase visualized in 2D may be one of the branches developed from the α -Fe phase [13]. Therefore, more details about these IMCs need to be determined.

To further observe the characteristic morphology of these IMCs mentioned above and identify their chemical compositions, SEM and EDS analyses are carried out on the IMCs in the typical microstructures of Al–Si–Mg cast alloys. Fig. 5 exhibits the typical SEM microstructures of Sr-modified Al–Si–Mg cast alloys with various Fe, Mn, and Mo additions. It indicates that the general features of these typical SEM microstructures show the α -Al dendrites (black), and eutectic Si particles (dark grey) morphology, consistent with those depicted by OM images (Fig. 4). Moreover, various IMCs (white, light grey or black) can be easily detected in the interdendritic regions. In general, IMCs can be carefully identified through the measured chemical composition combined with their characteristic morphology [12,36,38]. The element content results of some representative IMCs (points 1 to 8 in Fig. 5) determined by EDS point scanning are listed in Table 3. It can be inferred that the points 1, 2, 4, 6, 7, and 8 are Fe–IMCs, while the points 3 and 5 are Q-Al₅Cu₂Mg₈Si₆ and Mg₂Si, respectively. The observed Fe–IMCs are confirmed as α -Al₁₅(FeMn)₃Si₂, π -Al₈FeMg₃Si₆, β -Al₅FeSi, α -Al₁₅(FeMnMo)₃Si₂, α -Al₁₅(FeMo)₃Si₂, respectively. The above thermodynamic calculation results (Figs. 2 and 3) fully confirm this.

Usually, coarse polyhedral/blocky/star-like α -Al₁₅(FeMn)₃Si₂ phases and needle/plate-like β -Al₅FeSi phases might be responsible for a negative effect with regard to the mechanical performances of Al–Si–Mg cast alloys. Two key parameters, size and area fraction, are often selected to evaluate the Fe–IMCs [39,40]. The size and area fraction of Fe–IMCs in each alloy are measured and the quantitative results are listed in Fig. 6 and Table 4. Significantly, both the single addition of Mn or Mo, and the combined additions of Mn and Mo result in a great change in the size and area fraction of Fe–IMCs. The results indicate the decreased size and area fraction of α -Al₁₅(FeTM)₃Si₂ phases in Al–Si–Mg cast alloys with the decreasing Mn/Fe ratio (A1, A2, A3 alloys) or increasing Mo addition (A2, A4, A5, A6 alloys), agreeing with the description of the OM images (Fig. 4) above. Besides, both the maximum size and area fraction of needle/plate-like β -Al₅FeSi phases (in A3 alloy) also appear in this table. This result also confirms that, the Mn/Fe ratio can change both the type, morphology and size of Fe–IMCs, while the addition of Mo cannot change the type and morphology, but only the size [33].

3.3. Mechanical properties

To better evaluate the influences of Mn and Mo additions on the mechanical performances of Al–Si–Mg cast alloys with various Fe contents, hardness and tensile tests were performed, respectively. The results displayed below reflect the true potential of Al–Si–Mg cast alloys. It is very clear that, the hardness (Fig. 7) and tensile properties (Fig. 8) of Al–Si–Mg cast alloys are greatly improved by the appropriate Fe, Mn and Mo additions. On the one hand, without Mo addition, as the decrease of the Mn/Fe ratio, hardness, ultimate tensile strength (UTS), yield strength (YS) and elongation (EL) of the alloys both initially increase and subsequently decrease (A1, A2, A3 alloys), with their similar changing trends. Obviously, A2 alloy has the maximum hardness, UTS, YS, and EL values, reaching 78.24 HV, 243.41 MPa, 120.32 MPa, 5.76%, while those of A1 alloy (base alloy) are 75.06 HV, 232.30 MPa, 121.91 MPa, 4.12%, respectively. The observed results may be caused by the presence of brittle Fe–IMCs in the microstructure. The morphology and size of these Fe–IMCs have a profound effect on the mechanical performances [41]. In particular, the integrity of the Al matrix is easily lacerated by the needle/plate-like β -Al₅FeSi phase (in A3 alloy), and the larger size and area fraction of the α -Al₁₅(FeMn)₃Si₂ phase formed (in A1 alloy) at least as coarse and brittle as the β -Al₅FeSi phase

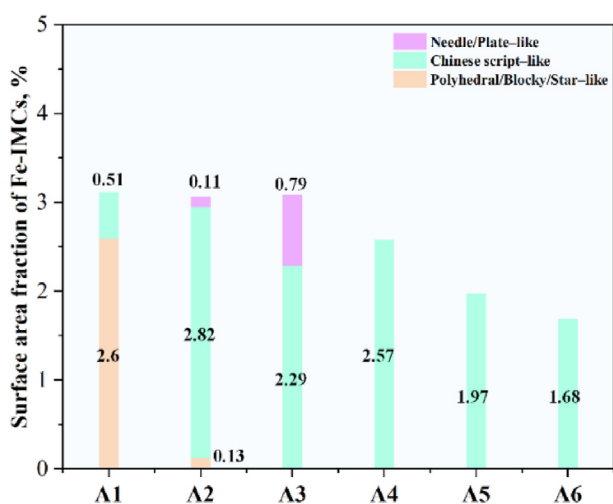
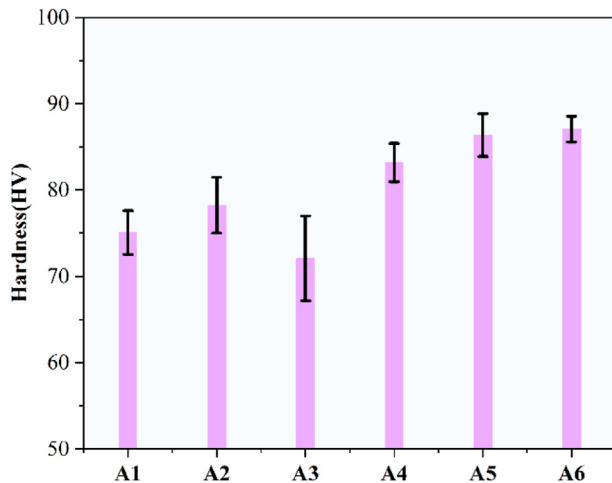


Fig. 6 – Surface area fraction of Fe–IMCs in Al–Si–Mg cast alloys with various Fe, Mn, and Mo additions.

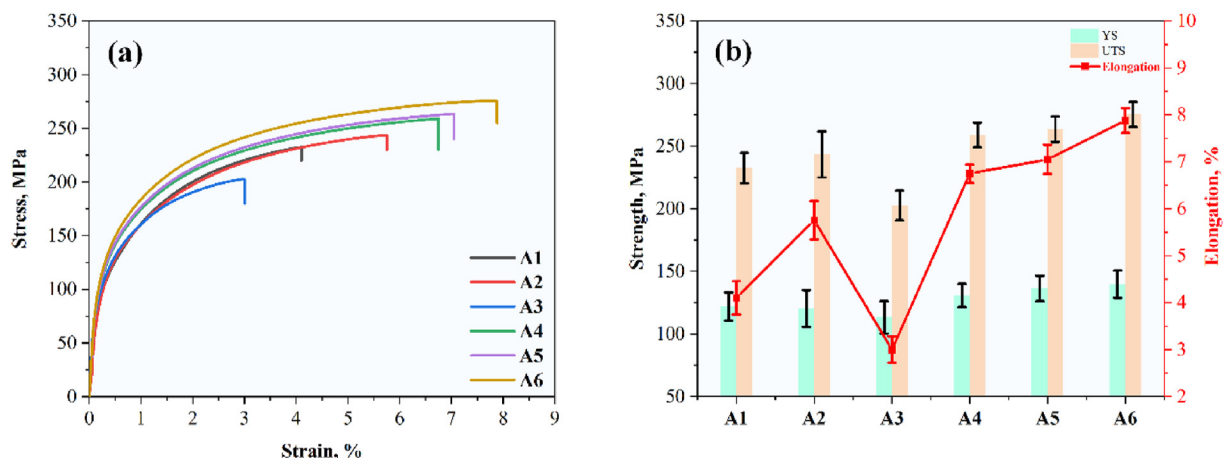
Table 4 – Quantitative results of the size and area fraction of Fe–IMCs.

Alloy	α -Al ₁₅ (FeTM) ₃ Si ₂			β -Al ₅ FeSi		
	Average length (μ m)	Average diameter (μ m)	Area fraction (%)	Average maximum length (μ m)	Average maximum width (μ m)	Area fraction (%)
A1	21.30 $\pm$ 9.19	10.36 $\pm$ 2.96	3.11 $\pm$ 1.26	—	—	—
A2	19.70 $\pm$ 8.20	9.61 $\pm$ 1.84	2.95 $\pm$ 1.07	14.24 $\pm$ 8.11	1.44 $\pm$ 0.32	0.11 $\pm$ 0.04
A3	15.25 $\pm$ 8.58	8.14 $\pm$ 2.12	2.29 $\pm$ 0.74	33.88 $\pm$ 10.37	2.55 $\pm$ 0.62	0.79 $\pm$ 0.27
A4	12.83 $\pm$ 6.95	6.97 $\pm$ 1.58	2.57 $\pm$ 1.15	—	—	—
A5	11.56 $\pm$ 4.95	6.47 $\pm$ 1.54	1.97 $\pm$ 0.98	—	—	—
A6	10.42 $\pm$ 5.48	4.02 $\pm$ 0.96	1.68 $\pm$ 0.83	—	—	—

**Fig. 7 – Micro–Vickers hardness histograms of α -Al dendrites in Al–Si–Mg cast alloys with different additions of Fe, Mn and Mo.**

[12,42]. However, the more compact Chinese script-like α -Al₁₅(FeMn)₃Si₂ phase (in A2 alloy) and α -Al are usually intertwined with each other, blocking the relatively strong splitting effect on the Al matrix [13,43]. In addition, the increased needle/plate-like β -Al₅FeSi phase and polyhedral/

blocky/star-like α -Al₁₅(FeMn)₃Si₂ phase are both detrimental to the fluidity of this alloy, resulting in more shrinkage cavity and porosity in both A1 and A3 alloys than the A2 alloy, and thus a sharp drop in the tensile strength and elongation of A1 and A3 alloys [38,44,45]. The results in Figs. 4–6 and Table 4 indirectly confirm this. It is noteworthy that, Seifeddine et al. [13] revealed the Mn levels affect the material strength mainly due to possible solid solution hardening. Obviously, the results here have little correlation with the possible solid solution hardening, since the mass fractions of Mn in the Al matrix are nearly the same, although the addition of Mn gradually increases in the alloys, as shown in Fig. 2(g–i). On the other hand, with the increase of Mo addition in Al–Si–Mg cast alloys, their hardness, UTS, YS, and El gradually increase (A2, A4, A5, A6 alloys), and their changing trends are also similar. In particular, the maximum hardness, UTS, YS, and El values of A6 alloy with 0.3 wt.% Mo addition are 87.04 HV, 275.34 MPa, 139.70 MPa, 7.88%, respectively, which are 11.25%, 13.12%, 16.11%, 36.81% higher than that of A2 alloy without Mo addition. The results clearly indicate that the gradual increase in UTS, YS, and El values of Al–Si–Mg cast alloys is closely associated with the decreased size and area fraction of Chinese script-like α -Al₁₅(FeTM)₃Si₂ phases, which has been fully demonstrated by the quantitative results of the size and area fraction of Fe–IMCs (in Fig. 6 and Table 4) [28]. In addition, the addition of Mo results in the disappearance of coarse and brittle α -Al₁₅(FeMn)₃Si₂ and π -Al₈FeMg₃Si₆, as well as needle/plate-like β -Al₅FeSi phases, thereby limiting the crack

**Fig. 8 – The tensile test results of Al–Si–Mg cast alloys with different additions of Fe, Mn and Mo. (a) Stress–strain curves; (b) tensile properties.**

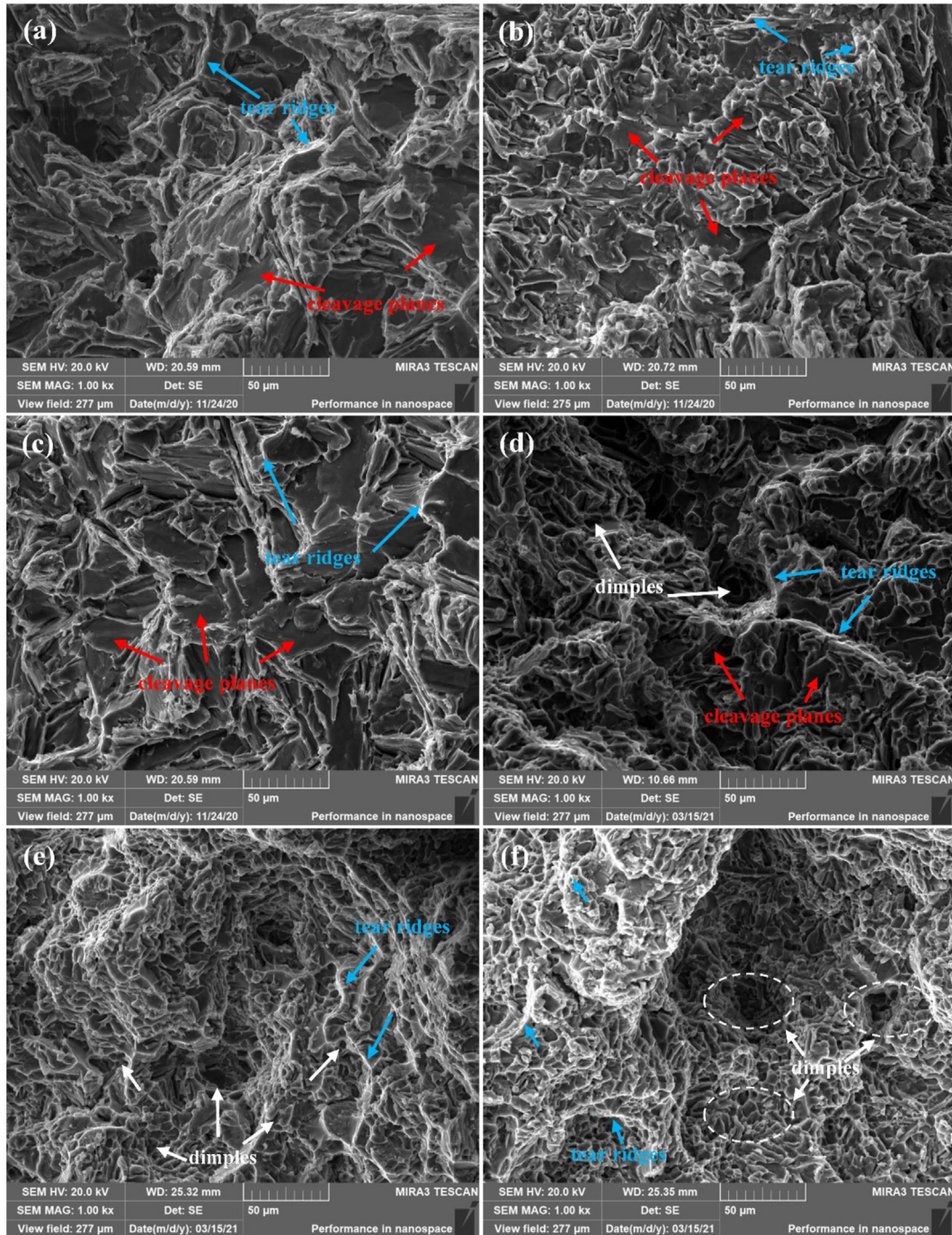


Fig. 9 – SEM micrographs of tensile fracture morphology of Al–Si–Mg cast alloys with different Fe, Mn, and Mo additions. (a) A1 alloy (base alloy); (b) A2 alloy; (c) A3 alloy; (d) A4 alloy; (e) A5 alloy; (f) A6 alloy.

initiation and propagation. Furthermore, the addition of Mo can also provide additional solid solution strengthening, which has been reflected in thermodynamic calculation results (Fig. 3(g–i)) and α -Al dendrites microhardness analyses (Fig. 7) [33].

Fig. 9 exhibits the tensile fracture morphologies of different Al–Si–Mg cast alloys investigated. It can be easily found from Fig. 9(a–c) that, these fracture surfaces are mainly covered by

irregular cleavage planes with few dimples. The fracture surface of A2 alloy has the smaller cleavage planes, while the larger cleavage planes (or cleavage steps) and tear ridges can be observed on the fracture surfaces of both A1 and A3 alloys. This demonstrates that the ductility of A1 and A3 alloys is worse than that of A2 alloy. Moreover, it can also be seen from Fig. 9(b, d–f) that, the fracture surface of A2 alloy without Mo addition is relatively flat, with small cleavage planes and

fracture steps. With the increase of Mo addition, the cleavage plane gradually decreases and the dimple number gradually increases (A4, A5, A6 alloys) on the fracture surfaces of Al–Si–Mg cast alloys. This further suggests that the ductility of this alloy can be greatly promoted by the addition of Mo. The fracture pattern gradually transforms from cleavage fractures to ductile fractures. Therefore, these tensile fracture morphologies exhibit a much more brittle failure mode for A1, A2, A3 alloys, while a more ductile failure mode can be obtained from A4, A5, A6 alloys. In general, the crack propagation and fracture processes of Al–Si–Mg cast alloys strongly depend on the morphology and size of eutectic Si phases and brittle Fe–IMCs [46,47]. Obviously, in Sr–modified Al–Si–Mg cast alloys, the fine eutectic Si particles are usually surrounded by a ductile Al matrix, inhibiting the damage progression to a certain extent and exhibiting high ductility. Therefore, it can be easily believed that the cracks in this study are mainly dominated by brittle Fe–IMCs rather than fine eutectic Si particles. The larger and more elongated the brittle Fe–IMCs are, the easier they are to split the Al matrix and thus fracture [48]. The above fractographic analyses demonstrate that, in Fig. 9(a–c), the fracture may be associated with the presence of a large number of fractured Fe–IMCs, whether the needle/plate-like β -Fe or polyhedral/blocky/star-like α -Fe phases, which can link and propagate the cracks, resulting in a premature failure. The limited ductility is significantly less than that when the Chinese script-like α -Fe phase is fractured, since the crack propagation of the latter may be more easily arrested by the surrounding Al matrix. Furthermore, the fracture mechanism seems to be the same for the alloys with the addition of Mo (Fig. 9(b, d–f)). With the increase of Mo addition, the ductility of the alloy is clearly increased, which is due to the decreased size and area fraction of Chinese script-like α -Fe phases, and the higher strength of the Al matrix. The results in Figs. 6 and 7 and Table 4 fully confirm this.

4. Conclusions

This investigation systematically evaluates the influences of Fe, Mn and Mo additions on the evolution of microstructure and mechanical performances of Al–Si–Mg cast alloys. The following conclusions can be drawn:

- (1) Based on experimental and thermodynamic calculations, the microstructures of Al–Si–Mg cast alloys are primarily composed of Al_3M (M means Ti, Zr, or TiZr), α -Al, eutectic Si, Mg_2Si , $Q-Al_5Cu_2Mg_8Si_6$, $\alpha-Al_{15}(-FeTM)_3Si_2$ ($TM = Mn, Mo$), $\beta-Al_5FeSi$ and $\pi-Al_8FeMg_3Si_6$.
- (2) The single addition of Mn facilitates the phase transformation from the polyhedral/blocky/star-like $\alpha-Al_{15}(FeMn)_3Si_2$ phase or needle/plate-like $\beta-Al_5FeSi$ phase to the Chinese script-like $\alpha-Al_{15}(FeMn)_3Si_2$ phase. Due to the formation of the Chinese script-like $\alpha-Al_{15}(FeMn)_3Si_2$ phase, the maximum hardness, ultimate tensile strength, yield strength and elongation of 78.24 HV, 243.41 MPa, 120.32 MPa, 5.76% of the A2 alloy, respectively.

- (3) The combined addition of Mn and Mo refines the Chinese script-like $\alpha-Al_{15}(FeMn)_3Si_2$ phases and promotes the formation of the Chinese script-like $\alpha-Al_{15}(-FeMnMo)_3Si_2$ phases. With the increase of Mo addition, $\beta-Al_5FeSi$, $\pi-Al_8FeMg_3Si_6$, and $\alpha-Al_{15}(FeMn)_3Si_2$ disappear one after another, the $\alpha-Al_{15}(FeMnMo)_3Si_2$ phase with smaller size and area fraction appear. In this article, maximum hardness, ultimate tensile strength, yield strength and elongation of 87.04 HV, 275.34 MPa, 139.70 MPa, 7.88% are obtained when Fe, Mn and Mo are added at 0.20wt.%, 0.02wt.%, and 0.27wt.%, respectively.
- (4) The improvement of mechanical properties, the crack propagation and fracture processes are closely related to the amount, size and morphology of Fe–IMCs. The polyhedral/blocky/star-like $\alpha-Al_{15}(FeMn)_3Si_2$ phase and the needle/plate-like $\beta-Al_5FeSi$ phase in Al–Si–Mg cast alloys with the additions of Fe, Mn, and Mo tend to produce stress concentration that leads to premature fracture of the alloy. However, the reduction of the size and area fraction of the Chinese script-like $\alpha-Al_{15}(-FeMnMo)_3Si_2$ phase is beneficial to improve the mechanical properties.
- (5) The work provides an efficient method to minimize or even eliminate the adverse effects of Fe–IMCs and thus improve the mechanical performances of Al–Si–Mg cast alloys, and offers a new idea for the extensive popularization of secondary Al–Si–Mg cast alloys in the lightweight design of automobiles.

Declaration of Competing Interest

The authors declare that they have no conflict of interest in this paper.

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